SAM D21 ARM Cortex-M0+ MCU Family - DATASHEET

Manufacturer: Microchip
Part Number: ATSAMD21G18A
Category: Microcontroller / ARM Cortex-M0+ MCU
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Overview

The Microchip ATSAMD21G18A is a 32-bit Arm Cortex-M0+ microcontroller running at up to 48 MHz with 256 KB Flash and 32 KB SRAM. Widely used in maker boards (Arduino Zero), IoT sensor nodes and wearables. Integrates 12-bit ADC, 10-bit DAC, USB 2.0 full-speed device/host and six SERCOM serial interfaces.

Peripheral Interfaces

6x SERCOM configurable as UART, SPI or I2C. USB 2.0 full-speed device and embedded host. 12-bit ADC at 350 ksps, up to 14 channels. 10-bit DAC at 350 ksps. 5x 16-bit timer/counters and 3x TCC. 32-bit RTC with calendar. 12-channel DMA. Supply: 1.62V to 3.63V. Package: 48-pin TQFP or QFN.

Key Specifications Summary

core: CortexM0+. freq: 48MHz. flash: 256KB. sram: 32KB. adc: 14ch_12bit. dac: 1x10bit. usb: device. sercom: 6.